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EE321: Computer Architecture
Control (1h)
2024 – 2025

Notes: Answer briefly and clearly using the provided space. No extra sheet will be accepted.

1. Z80 CPU Function and Interrupt Processing (20 pts)

Consider Z80 CPU with the following specifications: little-endian 8-bit processor, 16-bit address bus, a partial list of instructions and opcodes is given in Table 1, and the instruction formats are given in Table 2. The programming model is given in Table 3 and the GPR codes are given Table 4. The number of clock cycles for each Z80 machine cycle are given in Table 5. Consider the following HEX memory content:

2340H: 11 F1 12 F2 13 F3 14 F4 15 F5 16 F6 17 F7 18 F8
F610H: 01 F1 02 F2 03 F3 04 F4 05 F5 06 F6 07 F7 08 F8

1.1 Give the size of the registers A, MAR, MBR, and IR.

1.2 Give in decimal the 2'C 8-bit value stored in memory locations: a) 2348H and b) 2349H.

1.3 Give in decimal the 2'C 16-bit value stored in memory locations: a) 2348H and b) 2349H.

1.4 Give the size in bytes of the instructions: a) `LD IX,nn`, b) `LD HL,(nn)`, and c) `LD R,S`.

1.5 Give the HEX code of the instructions: a) `ADD A,-1`, b) `ADD A,(HL)`, c) `ADD A,H`.

1.6 Give the HEX content of the registers PC, MAR, and MBR after executing the **fetch** cycle of the instruction `LD IX,(2348H)` stored at address 2100H.

1.7 Give the HEX content of the registers PC, MAR, and MBR after executing the **execute** cycle of the instruction `LD IX,(2348H)` stored at address 2100H.

1.8 Which of the Z80 machine cycles (see Table 5) are involved in the instructions: a) `LD IX,nn`, b) `LD HL,(nn)`, and c) `LD R,S`?

1.9 Give the number of clock cycles (T-sates) it takes to execute (i.e. fetch and execute) the instructions: a) `LD IX,nn`, b) `LD HL,(nn)`, and c) `LD R,S`.

1.10 Assuming SP=F612H, give the HEX content of the registers SP, H, L, MAR, and MBR after execution the instruction `POP HL`.

1.11 We assume Z80 is configured in interrupt mode 2, register I=23H, and SP= F612H. The CPU receives an interrupt during the execution of the instruction `LD IX,0H` stored at address 2100H. The provided data by the interrupted device is 4CH. Give the HEX content of SP and PC **before** executing the **interrupt cycle.**

1.12 We assume the configuration of question 1.11, give the HEX content of SP and PC **after** executing the **interrupt cycle.**

1.13 We assume the configuration of question 1.11, give the HEX content of IR, MAR and MBR before fetching the first instruction of the ISR.

Instruction	Opcode (binary or HEX)	Instruction Size	Description
LD R, S	$01R_2R_1R_0\ S_2S_1S_0$		R= S
LD IX, nn	DD21H		IX = nn
LD IX, (nn)	DD2AH		IX = [nn]
LD HL, nn	21H		HL = nn
LD HL, (nn)	2AH		HL = [nn]
LD R, (HL)	$01\ R_2R_1R_0110$		R = [HL]
LD (HL), S	$01110\ S_2S_1S_0$		[HL] = S
LD (nn), A	32H		[nn] = A
PUSH IX	DDE5H		SP = SP-2, [SP] = IX
POP IX	DDE1H		IX=[SP], SP = SP+2
PUSH HL	E5H		SP = SP-2, [SP] = HL
POP HL	E1H		HL=[SP], SP = SP+2
ADD A, n	C6H		A = A + n
ADD A, (HL)	86H		A = A + [HL]
ADD A, S	$10000S_2S_1S_0$		A = A + S
INC (HL)	34H		[HL] = [HL] + 1
INC R	$00R_2R_1R_0100$		R = R + 1
HALT	76H		Stop the execution of the program

R =Destination register (it can be any GPR register).
S = Source register (it can be any GPR register).
$R_2R_1R_0$ = Binary code of the register R (see Table 4).
$S_2S_1S_0$ = Binary code of the register S (see Table 4).
n =8-bit value.
nn =16-bit value.
[…] = Content of memory location.

Table 1: Partial list of Z80 instructions.

1-byte
Opcode

1-byte instruction format

1-byte	1-byte
Opcode	Operand n or Opcode

2-bytes instruction format.

1-byte	2-bytes
Opcode	Operand nn or (Opcode + Operand n)

3-bytes instruction format.

2-bytes	2-bytes
Opcode	Operand nn

4-bytes instruction format

Table 2: Z80 instruction formats.

Register names	Description
A	Accumulator
B, C, D, E, H, and L	8-bit general purpose registers (GPR), can also be used in pairs as 16-bit registers (i.e. BC, DE, HL)
IX, IY	16-bit Address registers
F	8-bit flag register
I	8-bit interrupt register
R	8-bit refresh register
PC	Program counter
SP	Stack pointer

Table 3: Z80 programming model.

GPR	A	B	C	D	E	H	L	
Code	111	000	001	010	011	100	101	

Table 4: Binary codes of Z80 accumulator and GPR.

Z80 machine cycle	Number of clock cycles (T-states)
1-byte opcode fetch	4 clock cycles
1-byte operand fetch	3 clock cycles
1-byte memory read	3 clock cycles
1-byte memory write	3 clock cycles
1-byte I/O read	4 clock cycles
1-byte I/O write	4 clock cycles

Table 5: Number of clock cycle for each Z80 machine cycle.

Z80 interrupt signal	Mode	ISR location
Reset		0000H
NMI (Non-Maskable int.)		0066H
INT (Maskable int.)	0	ISR location can be one of the eight memory locations 00H, 08H, 10H, … 38H by using external hardware
	1	0038H
	2	8-bit data is supplied by the interrupting device. The ISR address is: [(I register * 256) + 8-bit data]

Table 6: Z80 interrupt processing.